



OpenPantry

Administrator Handbook

FOR ADMINISTRATORS

Setup, configuration, security, reporting, and day-to-day operation of OpenPantry.

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OVERVIEW

About OpenPantry

OpenPantry is a self-contained PHP + SQLite application that helps a food pantry check items out the door, keep a live inventory, and forecast what to reorder — with no database server, build step, or external dependencies to install. It is a hybrid Just-In-Time inventory system built for smaller pantries with limited storage and high-demand essentials.

Every outbound channel — grocery-style barcode checkout, the Menu Counter ordering kiosk, home deliveries, community events, and OrderAhead imports — feeds one inventory and one demand model, so the reorder report always sees the whole picture. This handbook is your reference for standing the system up, keeping it secure, and running it day to day.

WHAT YOU'RE RESPONSIBLE FOR

As administrator you own the four things volunteers never touch: the **OpenAI key & settings**, the **network / hours access gate**, the **admin password & encryption key backup**, and the **reorder cron job**. Each has its own section below.

GETTING INSTALLED

Requirements & deployment

- Any web host with **PHP 8+** and the **pdo_sqlite** extension (standard on every default install).
 - For field encryption, the PHP **sodium** extension (built in on PHP 7.2+).
 - The application folder (and **menucounter/**) must be **writable** by the web server so the databases and encryption key can be created on first hit.
- 1 Copy the whole OpenPantry folder to your host. The nested Menu Counter app ships inside it — no separate deploy.
 - 2 Browse to the app root. The schema builds itself, the produce table seeds, and **openpantry.db** plus **menucounter/picklist.db** are created automatically.
 - 3 Open **Settings** and complete first-time configuration (next section).

DO THIS BEFORE OPENING

First-time configuration (Settings)

Everything below lives on the **Settings** page:

- **Administrator password** — change it from the default **admin** immediately. It is stored one-way hashed and can't be recovered, only reset. Changing it later requires entering the **current password** first, so a walk-up at an unattended screen (or a stolen login cookie) can't silently swap it and lock everyone out.

- **Supervisor password** (optional) — a second login for whoever opens the pantry. It reaches every screen the administrator password does, except that on **Settings** it can only change the **Public IPv4 Address** — the rest of the page is read-only. That lets a shift lead re-point the network gate when the pantry's address changes without being handed the OpenAI key, the mail settings, or the passwords. Setting or removing it asks for the administrator password. Leave it unset and the role doesn't exist — there is no default supervisor password.
- **OpenAI API key** — powers automatic brand→generic naming on new barcodes. Use the **test** button to confirm it works. Stored encrypted at rest.
- **Network Access IP** — set to your pantry's public Wi-Fi address so the kiosks only work on-site. Leave blank during setup to avoid locking yourself out, then set it.
- **Allowed hours** — optional weekly schedule that closes the kiosks outside service times.
- **Administrator email** — where reorder-reminder digests are sent — and where login soft-lock codes go (next section), so keep it current.
- **Email notifications (SMTP)** — optional authenticated sending; otherwise the app uses PHP `mail()`. Use **Send Test Email** to verify delivery.
- **Produce tare** — ounces subtracted from hand-typed produce weights (not scale readings).
- **USB scale pairing** — a HID Point-of-Sale scale (the kind USPS postage software uses, e.g. the DYMO M25) plugs straight into the station with no adapter or driver. Open the scanning page in Chrome or Edge, click **Connect Scale** once and pick the device; the browser remembers the grant, so every later visit reconnects on its own. Do this once per station and per browser profile — there is no app setting for it. A station still using the older cable-connected scale needs none of this: it types weights in as a keyboard would, and the scale strip described in the Volunteer Handbook never appears.
- **Par-level defaults** — default lead time and safety-stock Z used by the Order Now report.
- **Food pantry name & logo** — branding shown in the header and on printed sheets.

PROTECT THIS ABOVE ALL

Security & data privacy

- **Admin password** is stored as a one-way hash (`password_hash`); even the running app can't read it back. The login cookie is derived from the hash, so changing the password instantly logs everyone out.
- **Supervisor password**, when set, is hashed and cookie-bound exactly like the admin one, and the cookie records which of the two opened the session — so a supervisor session stays a supervisor session, and changing or removing that password signs out everyone holding it.
- **Login rate limiting** throttles failed admin logins per IP: two free retries, then 10- and 30-second waits; at the fifth failure the login **soft-locks** and a single-use 6-digit code is emailed to the administrator address (with an escalating 1–10 minute timeout as the fallback when no code can be sent). A successful login clears the slate. Covers both the OpenPantry login and the Menu Counter item admin.
- **Network + hours gate** (in `auth.php`) blocks any device that isn't on the allowed IP or is outside allowed hours, with a styled “Access Denied” wall.
- **Encryption at rest** (libsodium) now covers **every Settings value** — the OpenAI key, allowed IP, SMTP credentials, and the rest (only the already-hashed admin and supervisor passwords stay hashes) — plus delivery clients' address / city / phone. Existing plaintext rows are upgraded automatically on the next load.

KEEP THE DATABASES OFF THE WEB

The root `.htaccess` refuses to serve `*.db`, `*.db-wal`, `*.db-shm`, and `*.sqlite` files over HTTP, so the databases can't be downloaded by URL even on a default install. For defense in depth, point the `OPENPANTRY_DB_DIR` SetEnv (in `.htaccess`, alongside `OPENPANTRY_KEY_PATH`) at a directory **outside public_html** — both databases are stored there, and an existing in-webroot database is checkpointed and moved across automatically on the next request. The SetEnv values are also honored on CLI runs, so the cron mailer resolves the same files.

THREE TIERS OF ACCESS — AND WHY VOLUNTEERS DON'T LOG IN

The **volunteer kiosks are gated by the allowed IP only**: the scanning stations, the Menu Counter order form and pick queue, and the delivery kiosk open straight to the work screen on the pantry network, with **no password**. The **admin password guards the administrative screens** — the dashboard, Settings, Inventory, Restock, Lookup Tables, Reports, the Menu Counter item admin, and the delivery client roster. The optional **supervisor password** sits between the two: the same administrative screens, but Settings is read-only apart from the IP address controls. Setting the allowed IP correctly is therefore what keeps the kiosks safe, so keep it current whenever the pantry's public address changes.

BACK UP ENCRYPTION_KEY.PHP

The 32-byte key in `encryption_key.php` is generated on first use and is the **only** thing that can decrypt your data. If you lose it, all encrypted fields are gone for good. Back it up, keep it out of version control, and ideally relocate it above the web root via the `OPENPANTRY_KEY_PATH` environment variable or a `FS_ENC_KEY_PATH` constant. On PHP without libsodium, encryption silently degrades to plaintext until a sodium-capable PHP runs.

KEEPING COUNTS HONEST

Managing inventory

- **Inventory page** — the canonical current-count list. Create items here and set each one's unit (`each` or `lb`).
- **Restock** — stage counts and submit a batch that *adds* to inventory, flagged purchased vs. donated (drives the Purchased-% column). No order rows are written, so restocks stay out of usage reports.
- **Deliverable flag** — uncheck it to keep an item in inventory but hide it from the Menu Counter order form.
- **Count-per-case** — set how many units a supplier case holds; the Order Now report then shows a Case Request column (order need ÷ case size, rounded up).
- **Order unit & Avg Wt** — for produce the pantry weighs but the vendor sells by the piece (avocados on the scale in `lb`, bought as a 48-count case): set Order Unit to `each` and Avg Wt to the average pounds per piece. Count-per-case is then read in the order unit, the order sheet and email are written in it, and Restock Now converts the delivery back to pounds. Leave Order Unit on “same as stock” for everything else. Without an Avg Wt there is nothing to convert by, so the order falls back to the stock unit.
- **Checkout, deliveries, events, OrderAhead** all decrement inventory automatically as orders close or imports run.

BARCODES TO GENERIC NAMES

Lookup tables & naming

OpenPantry stores each item by a plain **generic name** (e.g. “Black Beans”) rather than brand, so demand aggregates cleanly. The first time a UPC is scanned, the app looks it up on Open Food Facts and asks OpenAI to reduce it to a 2–4 word generic, then caches the mapping.

- **Produce** — PLU codes (and 12-digit pantry labels starting with 4) map to produce names; the table ships pre-seeded and you can add more.
- **UPC / Generic Cache** — review and edit AI-derived names in place. If it guessed “Beans” when you wanted “Black Beans”, fix it here and future scans use your edit.
- **Add UPC Manually** — for codes Open Food Facts can't resolve: type the UPC, an optional branded name, and the generic name, and the mapping is cached (source [manual](#)) for all future scans of that code.

Merging duplicate item names

The generic name *is* the item — there is no id behind it — so the same food can end up stored under two spellings. The AI returns “Green beans” one week and “Green Beans” the next; a produce code says “Squash Zucchini” while a UPC says “Zucchini”; someone retypes a name. Each variant then keeps its own scan history, its own inventory row, and its own reorder alert, which splits demand in half and **biases both par levels low**. The **De-duplicate** tool merges them back together. It is deliberately kept off the menu — browse to </openpantry/deduplicate/> when you need it.

- **Possible Duplicates** — names that match once case, punctuation, word order, and plurals are ignored are grouped for you. Click **keep** on the spelling you want and **merge** on the one to fold into it.
- **What a merge moves** — every scan (so the demand history recombines), the UPC mappings and produce codes (so future scans land on the kept name), the inventory row (counts and lifetime restock totals are summed; the kept row's unit, case size, and Order Unit settings win), and any reorder alert. The duplicate then no longer exists.
- **Renaming an item outright** — type a name in the **or type a new name** box instead of picking one, and the whole item moves to that spelling everywhere.
- **“No barcode”** in the name table means no UPC or produce code still resolves to that name, so nothing new can be scanned into it. Usually a leftover from a rename — a good merge candidate.

A MERGE REWRITES HISTORY AND CANNOT BE UNDONE

Merging edits past scans, not just future ones — that is the point, but there is no undo. Back up [openpantry.db](#) before a big cleanup session. The reorder report needs no attention afterward: forecast fits are keyed to each item's scan history, so a merged item refits by itself on the next report load.

MERGE BEFORE YOU TRUST A NEW ITEM'S PAR LEVEL

Duplicates are worth a look whenever an item you know is moving shows a surprisingly low order need — a second spelling is quietly holding the other half of its scans. The Menu Counter has its own separate de-duplicate tool for the order-form items in its own database.

TURNING SCANS INTO DECISIONS

Reports & the demand model

- **Order Now** — the reorder report. Computes a Par Level and how much to order per item.
- **Orders Listing** — every order and its items over a date range (delivery/event orders are tagged).
- **Item Usage** — per-item totals over a date range.
- **Daily Volume** — orders and scans per day.
- **Basket Size** — distribution of items per in-pantry trip over time.
- **Impact** — the report you hand a board member or a grant officer: pounds distributed, households and people reached, top items, and where the food came from, over a window that defaults to the last 12 months. It is the only report that reads **both** databases — scans and orders from `openpantry.db`, counter requests and household sizes from `picklist.db` — and it keeps the two in separate sections rather than summing them, because a request a household typed and food that left the building are different events.

THE IMPACT REPORT'S TWO ASSUMPTIONS ARE YOURS TO SET

Packaged goods are counted, not weighed, so any pound figure covering them rests on an assumed **average pound per packaged item**; the meals-equivalent figure rests on an assumed **pounds per meal** (Feeding America uses 1.2). Both sit at the top of the report as editable inputs and are restated in its **Methodology & Caveats** card, so anyone reading the headline numbers can see what they depend on. Set them to whatever your funder expects before you print.

How Par Level is computed

For each item: **Par Level = Forecast(LeadTime) + SafetyStock**, where **SafetyStock = $Z \times \sqrt{\text{Variance(LeadTime)}}$** and **Order need = max(0, Par Level – latest inventory count)**. Forecast and variance come from a quasi-Poisson model fitted to each item's weekly scan history, with a time trend and annual seasonality; dispersion inflates the safety stock so it reflects real volatility. Z defaults to 1.65 (~95%). Items with under ~2 months of history fall back to a trailing average and are marked with a °.

THE FORECAST CACHE IS DISPOSABLE

Fits are memoized in the `forecast_cache` table and rebuild on demand. If a report ever looks stale or wrong, that table is safe to delete — it will rebuild from the scan history.

Reorder alerts & email

Set a per-item lead-time alert and it shows as a banner on Order Now when projected days-of-stock drop below the threshold. Tick its **Email** box to also have it emailed. The cron job (`cron_reorder_alerts.php`) mails a digest of triggered, email-flagged items to the administrator address, at most once per ~20 hours per item.

AUTOMATED REMINDERS

The reorder cron job

Add a cron job (cPanel → Cron Jobs) that runs the mailer on your cadence, e.g. daily at 7am:

```
0 7 * * * /usr/local/bin/php
/home/you/public_html/openpantry/cron_reorder_alerts.php
```

CALL PHP BY ITS ABSOLUTE PATH

Use the full path to the PHP binary (e.g. `/usr/local/bin/php` on Namecheap-style cPanel hosts). A bare `.php` path can fail silently with “Permission denied” and no email ever goes out. Confirm with **Send Test Email** in Settings first.

THE ORDERING KIOSK

Administering the Menu Counter

- **Item admin** — add, remove, reorder (drag & drop), and toggle items; set category, name, sizes, and the **Family Factor** (multiplied by household size, capped at 5, rounded up) that decides how many units land in the pick queue.
- **Pick queue (Orders)** — the live back-of-house dashboard volunteers pull from; auto-refreshes every 30 seconds.
- **Deduplicate** — merge duplicate item rows that crept in over time. This one is the Menu Counter's own order-form items; the item-name merge tool under Lookup Tables is a separate page for a separate database.
- **Reports** — item-usage reports with a chart; shoppers are anonymized as “Client N.”
- The Menu Counter shares the same login, allowed-IP, and allowed-hours gate as the rest of OpenPantry. Its order form shows exactly the items toggled **On** in Item admin — the Inventory page's Deliverable checkbox only affects the delivery menu, not the Menu Counter.

HOME DELIVERIES

Delivery paperwork & the AI upload

- **Print Menus (order forms)** — one page per pending client, with a checkbox per in-stock item. Each item shows its **default Qty / Weight in parentheses** — e.g. **(2 each)** or **(1.5 lb)** — computed from that client's household size with the same rules the packing list uses, so the two sheets always agree. A bold footer on every page asks volunteers to write changes next to the default amount only when the client requests them.
- **Packing & Delivery Lists** — per-client pick sheets for orders recorded this round. Count items show as **2 each**; produce delivered by weight shows as **1.5 lb**.
- **AI upload** — completed forms are scanned to one PDF and read by vision AI in two independent passes; disagreements are flagged “verify against the paper” in the results table.

THE AI READS CHECKBOXES ONLY

The upload reader detects which boxes are marked — it does **not** read handwritten Qty / Weight corrections. When a volunteer writes a changed amount next to an item (per the footer instruction), apply that change by hand when packing; the packing list will otherwise show the standard computed amount.

KEEPING IT HEALTHY

Maintenance & troubleshooting

- **Backups** — back up `openpantry.db`, `menucounter/picklist.db`, and `encryption_key.php` together. The two databases hold your data; the key decrypts it. If you've set `OPENPANTRY_DB_DIR`, both databases live in that private folder instead of the app folders.
- **No email arriving?** Verify the admin email and SMTP in Settings, send a test, and check the cron PHP path (above). Without SMTP the app falls back to PHP `mail()`.
- **New barcode saved with a raw name?** The OpenAI step was skipped (bad/rate-limited key). Fix the key and edit the name under Lookup Tables.
- **Same item listed twice in a report?** Two spellings of one generic name. Merge them at `/openpantry/deduplicate/`, which also recombines the split scan history the par levels are built from.
- **“Access Denied” for legitimate staff?** The public Wi-Fi IP changed or you're outside allowed hours. Update the allowed IP in Settings.
- **Locked out by the login throttle?** Wait out the timer, or use the 6-digit code emailed to the administrator address. A successful login resets the counter; 30 quiet minutes decay it on their own.
- **Report looks off?** Delete the `forecast_cache` table; it rebuilds from scans.

POST THESE AT EACH STATION

Opening checklists (for training)

These are the same checklists in the Volunteer Handbook. Print and post one at each station, and use them when training new volunteers.

CHECKOUT · LASER SCANNER OR CAMERA

Open-the-Station Checklist — Scanning Station

Work top to bottom. Tick each box as you go.

☐

Power on the Chromebook tablet and connect it to the pantry Wi-Fi (the station only works on the pantry network) if it is not already connected.

☐

Make sure the handheld barcode scanner is connected to the tablet; wait for its ready beep / steady LED.

☐

Power on the produce scale and confirm the scale strip reads **Scale ready**; see the setup below if it does not.

☐

Open the browser and go to the OpenPantry **Scan** page. It opens straight to the scanner — **no password** is needed, because the pantry network authorizes the station.

☐

Scan any item's barcode as a test — a “Last scan” line should appear. Then tap the red **×** **Cancel Order** so the test doesn't count.

☐

Tap once inside the barcode box so the cursor sits there (it glows green when focused).

NO WIRED SCANNER? USE THE CAMERA ON THE SAME PAGE

There is one **Scan** page for both ways of scanning. On a phone or any tablet without a handheld scanner, tap **Start Camera** in the *This Order* box and point the camera at the barcode — the page chirps once for each code it reads. **Stop Camera** puts it away. The camera needs permission the first time, and the page must be opened over **https**.

CHECKOUT · THE PRODUCE SCALE

Setting up the produce scale (once per tablet)

The **DYMO M25** is a USB scale. It plugs into the tablet and sends its readings to OpenPantry on its own — there is no keypad sequence to enter, no mode to select, and no unit to set, because the station converts whatever unit the scale displays into pounds. All it needs is power and, the first time on a given tablet, the browser's permission:

1	Stand the scale on a flat, dry surface with nothing on the platform.	Nothing on the platform
2	Plug its USB cable into the tablet.	Connected to the station
3	Press the scale's power key and let the display settle to 0 .	Zero / stable reading
4	On the Scan page, read the scale strip just above the barcode box.	Says what the scale is doing
5	If it says Scale not connected , tap  Connect Scale and pick the scale from the browser's list.	Once per tablet only
6	Confirm the strip reads  Scale ready .	Interfaced to OpenPantry

WHEN THE STRIP SAYS “SCALE READY”, YOU'RE SET

That line means the station is hearing the scale. You can then weigh produce in **either order** — scan the PLU and set the item on the scale, or weigh it first and scan the PLU when the station asks for it. You only ever tap **Connect Scale** once on a tablet: the browser remembers the scale and reconnects by itself every later shift.

THE SCALE SWITCHES ITSELF OFF

The M25 powers down after a quiet spell to save its batteries, and a scale that has shut off looks exactly like one nobody has touched. If the strip says  **Make sure scale is on.**, press the scale's power key to wake it — you do *not* have to redo the steps above. If the display is awake but not reading **0** with an empty platform, press its **Tare / Zero** key.

CUSTOMER ORDERING KIOSK

Open-the-Station Checklist — Menu Counter

The Menu Counter has two screens: the **customer order form** that shoppers tap, and the **pick queue** that volunteers pull from in the back.

Customer-facing tablet

- ☐ Power on the customer tablet and connect it to the pantry Wi-Fi.
- ☐ Open the browser to the OpenPantry **Menu Counter** order form. It opens directly — **no password** to enter.
- ☐ Confirm the page loads the item buttons grouped by category (Dairy, Dry Goods, Frozen, ...).
- ☐ Set the language selector back to English so the next shopper starts fresh.
- ☐ Stand the tablet in its holder facing the shopper; clean the screen.

Pick-queue tablet / screen (back of house)

- ☐ Open the **Menu Counter** → **Orders** (pick queue) page on the volunteer screen. It opens directly — **no password** needed.
- ☐ Confirm the queue auto-refreshes and the topbar shows “pending orders” and “items remaining”.
- ☐ Submit one test order from the customer tablet and confirm it appears in the queue, then Mark Complete to clear it.
- ☐ Make sure bags/bins and a pen are staged at the pick bench.

NO LOGIN ON THE PANTRY NETWORK

The Menu Counter order form and pick queue are secured by the pantry's network address, not a password, so volunteers never log in to use them. A password is only asked for on the administrator's **item-setup** screen — leave that to the person opening the pantry.